# Devansh Mayatra

**MS Applied Machine Learning | Machine Learning, Deep Learning & Production Systems**
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## Education

**University of Maryland, College Park** — **College Park, MD**
*Master of Science in Applied Machine Learning* — *Expected May 2028*

**Vasantdada Patil Pratishthan's College of Engineering, University of Mumbai** — **Mumbai, India**
*Bachelor of Engineering in Computer Engineering | CGPA: 7.95/10.0* — *May 2026*

## Experience

**Robin Home AI** — **Mumbai, India**
*Full Stack Developer Intern* — *Aug 2025 – May 2026*

- Built core mobile and backend workflows for Robin Home AI's first beta release within a 6-month development cycle, working directly with the CEO and CTO on product and engineering decisions; the product later scaled to **1M+ Google Play downloads**.
- Developed Python preprocessing for telemetry/sensor logs and asynchronous FastAPI services backed by Redis, PostgreSQL, EC2, RDS, and S3, keeping key backend workflows at **<100 ms** response time.
- Instrumented onboarding, messaging, and call-forwarding flows with product analytics and event tracking, giving the team visibility into user behavior and feature drop-off across the beta experience.

**Kraft Nexus** — **Mumbai, India**
*Junior React JS Developer* — *Apr 2025 – Aug 2025*

- Consolidated repeated frontend patterns into **30+** reusable React components shared across production applications, saving about **15 engineering hours/week**.
- Optimized rendering and page behavior across production interfaces, improving responsiveness and raising Lighthouse scores by **15 points**.
- Built data-heavy aviation interfaces for training status, flight-duty limits, calendar views, and operational charts, translating domain rules into maintainable frontend workflows.

**JKB Education and IT Services** — **Mumbai, India**
*Full-Stack Developer Intern* — *Nov 2023 – Mar 2025*

- Replaced manual scoring and reporting with full-stack prediction and operations tools used by roughly **100–200 daily users**.
- Automated repetitive reporting and administrative workflows, reducing manual operational effort by **80%+**.
- Delivered and maintained web/mobile features across client projects, connecting user-facing workflows to backend services and databases over a **16-month** engagement.

## Projects

**Neural Networks from First Principles – PyTorch, MLPs & Backpropagation** **[GitHub]**

- Implemented a **27-class** character-level neural network with learned embeddings, fan-in-aware initialization, BatchNorm, tanh, numerically stable softmax, and cross-entropy to study model behavior below framework abstractions.
- Manually derived gradients for **26** tensors/parameters and matched PyTorch autograd to $2.24 \times 10^{-8}$ maximum absolute deviation, using activation/gradient statistics to diagnose saturation and optimization behavior.
- Extended the same first-principles workflow into computer-vision fundamentals, implementing material through convolutional-network concepts while studying modern CNN architectures and residual learning.

**DocuMind AI – Retrieval-Augmented Generation** **[Demo] [GitHub]**

- Built multi-format PDF/DOCX/TXT/Markdown ingestion with Qdrant, dense + BM42 hybrid retrieval, and FlashRank reranking to handle cases where naive semantic search missed relevant context.
- Added query reformulation, source attribution, and streamed generation so answers remain traceable to retrieved evidence instead of behaving like an ungrounded chatbot.
- Deployed an interactive Streamlit application with multi-format document support, turning the retrieval pipeline into a usable end-to-end system rather than a notebook-only prototype.

**SmartDispatch AI – End-to-End Machine Learning Pipeline** **[GitHub]**

- Engineered cyclic temporal features and spatial K-Means clusters to capture time- and location-dependent demand patterns before training LightGBM models for price prediction.
- Evaluated the learned model against the project's baseline and achieved a **90.5% improvement** on the tracked prediction metric.
- Packaged inference behind FastAPI and Docker, separating training from serving so experiments could be reproduced and predictions exposed through a production-style API.

## Technical Skills

**Languages:** Python, C++, Go, TypeScript, JavaScript, SQL, Bash/POSIX shell
**ML / DL:** PyTorch, NumPy, pandas, scikit-learn, LightGBM, XGBoost; neural networks, backpropagation, BatchNorm, initialization, optimization, CNNs, convolution/pooling, clustering, model evaluation
**Backend / Web:** FastAPI, Flask, React, Next.js, Node.js, REST APIs, async programming; PostgreSQL, Redis, MongoDB, Qdrant
**Systems / Cloud:** Docker, Linux, Git, CI/CD; AWS (S3, EC2, RDS)